

## REVIEW

# AI4Protein: transforming the future of protein design

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Artificial intelligence (AI) has revolutionized the protein engineering process from multiple aspects, including representing protein information, generating protein designs, and evaluating protein properties. This review aims to introduce the recent progress of AI in protein research. We first introduce how AI models represent protein sequences, structures, and other properties. Further, the applications of generative models in protein design are introduced. The use of predictive models and smart agents in the evaluation process is then discussed, including high-precision protein property simulation and wet lab experimental design. Additionally, we discuss the future development of AI in protein research and the potential challenges it may encounter.

protein engineering | protein design | artificial intelligence

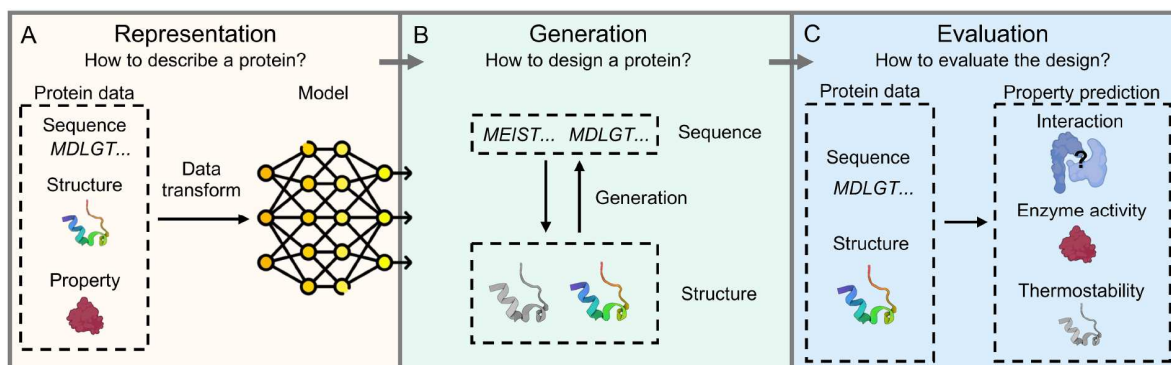
## Introduction

Recently, the Nobel Prize in Chemistry was awarded to pioneers in the field of computational protein science, recognizing their groundbreaking contributions to protein structure prediction and design. As key innovations in this field, tools like RoseTTA-Fold (Baek et al., 2021), AlphaFold2 (Jumper et al., 2021), RoseTTAFold All-Atom (Krishna et al., 2024), and AlphaFold3 (Abramson et al., 2024) demonstrate the transformative power of artificial intelligence (AI) in revolutionizing the protein engineering field. These achievements underscore the growing importance of AI in accelerating and enhancing the protein design process, enabling researchers to explore new frontiers in biology and medicine that were previously unattainable through traditional methods. The rapid advancements in AI have introduced a new era in protein research, addressing some of the most challenging bottlenecks that traditional experimental methods face. AI can swiftly process vast amounts of biological data, automating labor-intensive tasks that typically require significant time and resources. It excels at predicting protein structures, simulating interactions, and even generating novel protein sequences tailored to specific functional needs. These capabilities reduce reliance on time-consuming lab experiments and offer an unprecedented level of precision and scalability in protein engineering. For example, typical protein engineering processes heavily rely on structural information. Researchers collect data from experiments using techniques like X-ray crystallography and Nuclear Magnetic Resonance (NMR) spectroscopy, analyze these sequences, and iteratively test new designs in the lab through mutagenesis or directed evolution. This process involves generating new variants, performing

structural and functional tests, and refining the designs over multiple experimental cycles. While some of these steps can be enhanced or even replaced by advanced AI tools, such as the automated construction of atomic models and the identification of proteins in cryo-EM maps using ModelAngelo (Jamali et al., 2024), the trial-and-error nature of this approach, although effective, still remains time-consuming and resource-intensive.

In this review, we introduce how AI has facilitated the protein design workflow, with a focus on streamlining the processes of design, analysis, and validation. The process can be divided into three main stages: representation (Figure 1A), generation (Figure 1B), and evaluation (Figure 1C). In the representation stage (Figure 1A), we discuss how AI systems transform raw protein data, including sequences, structures, and properties, into representations that models can efficiently process and analyze. These representations are more easily processed by models, enabling more efficient analysis. For example, sequence data might be converted into vector-based representations that capture critical biological features, while structural data could be encoded into formats that facilitate predictive modeling. For the generation stage (Figure 1B), we explore AI's role in creating novel protein designs, where generative models propose new sequences and structures optimized for specific functions or properties, such as enhanced binding affinity or improved stability. These models leverage large-scale data to explore design spaces that may be difficult or time-consuming to discover through traditional methods. Finally, in the evaluation section (Figure 1C), we focus on how AI-driven models help evaluate protein properties by prediction, including interactions and functionality such as enzyme activities and thermostability, thus accelerating the validation process and minimizing the reliance

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**Figure 1.** Key stages where AI enhances the protein engineering workflow. A, AI models convert raw protein data such as sequences, structures, and properties into usable formats for later computation. B, AI-driven models design novel protein sequences and structures with tailored properties. C, AI models can make property predictions, such as interactional and functional outcomes of designed proteins, significantly accelerating the validation process compared with traditional experimental methods.

on traditional, labor-intensive experimental methods. Through these three stages, we highlight how AI is reshaping the landscape of protein engineering, making it faster, more scalable, and more precise. The AI technology holds vast potential for further development in the protein engineering field. The algorithm design and the collection and use of protein data continue to present valuable research opportunities. Moreover, while AI accelerates many stages of the design process, protein designs ultimately require experimental validation. This underscores the importance of exploring wet-lab automation, which presents both significant challenges and exciting opportunities for the future of protein research. By reflecting on these innovations and looking ahead, we discuss both the current impact and the vast potential for future growth, especially in algorithm development and wet-lab automation.

## Representing protein data

The first step to employing AI for protein research is to transform biological information into forms that AI models can process. Protein sequences determine protein structures, which further determine the biological functions. To build the connections between these properties, the sequential and structural information is first transformed into languages that AI can understand, which is termed as representation. In this section, we explore how effective sequence, structure, and property modeling can enhance the representation of protein knowledge in AI models, enabling more accurate predictions and providing more insights in protein research (Figure 2).

### Sequence representation

Protein sequences are fundamental to the biochemical functions and life activities of organisms, as they determine the structure and activity of proteins essential for metabolism, signaling, and cellular processes. The dependencies between amino acids within these sequences are critical for determining how a protein folds, stabilizes, and interacts with other molecules. Therefore, the comprehension of these dependencies is essential for enhancing model predictive and analytical capabilities.

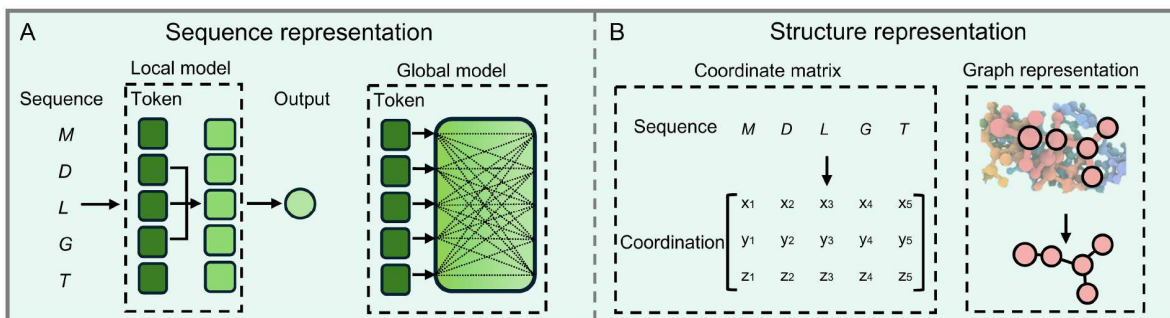
AI models usually convert discrete sequences of amino acids into continuous vector representations (i.e., tokens). These tokens map sequences into a high-dimensional space, where similar parts of protein sequences are positioned closer together,

reflecting their structural or functional relationships. Local-dependency models effectively capture and represent important sequence information, allowing AI models to identify patterns within the sequence. However, they face limitations when it comes to modeling long-range or more complex dependencies between distant parts of the sequence, which are often essential for understanding protein folding and function. Global-dependency models overcome these challenges by capturing both local and long-range relationships, providing a more detailed understanding of how sequence variations can impact protein structure and function. In the following sections, we will explore these two approaches in more detail, highlighting their respective strengths and limitations in protein sequence modeling (Figure 2A).

### Local dependency

The relationships between neighboring amino acids in a protein sequence are crucial for determining local structural features such as alpha-helices and beta-sheets. Consequently, AI models designed to capture such local dependencies have become essential tools in protein research. These models usually capture local dependency by focusing on short, contextually relevant windows within a sequence, allowing them to learn relationships between neighboring amino acids that contribute to the formation of local structural features.

An early study, ProtVec (Asgari et al., 2019), introduces the first vector-based representation of protein sequences. Inspired by Word2Vec (Mikolov et al., 2013), it captures local dependencies by splitting protein sequences into overlapping 3-grams (segments of three adjacent amino acids) and using a neural network to map these short segments into a continuous vector space. By focusing on these small, localized segments, ProtVec effectively models short-range interactions, positioning sequences with similar local patterns closer together in the vector space. Later, instead of representing overlapping 3-grams of sequences, Kimothi et al. (2016) apply Doc2Vec (Le and Mikolov, 2014) to represent entire protein sequences as single vectors. ProtVecX builds on ProtVec by representing protein sequences with variable-length sub-sequences instead of fixed 3-grams (Asgari et al., 2019), offering more flexibility and capturing richer information. SeqVec (Heinzinger et al., 2019) introduces a significant advancement by utilizing ELMo (Peters et al., 2018), which captures local dependencies through a bidirectional language model that considers both preceding and following



**Figure 2.** The mapping relation of biological molecules and AI model representations. A, Protein sequences can be represented into tokens and then processed with local dependency and global dependency models. Local dependency models could capture important local sequence information, while global dependency models can further understand long-range relationships within sequences. B, Protein structure data can be represented as coordinate matrix and graph.

amino acids. This approach models short-range interactions within sequences by using a sliding context window, allowing for more complex relationships to be learned and leading to more accurate predictions of protein structure and function. As a crucial neural network model, convolutional neural networks (CNNs) effectively capture local dependencies by applying convolutional kernels to fixed-sized windows of protein sequences, focusing on short-range interactions between neighboring amino acids. For instance, Öztürk et al. (2018) represent the protein sequence with numerical labels to predict drug-target binding affinity. However, despite these methods being successful in modeling local dependencies, they often struggle to capture long-range interactions, which are essential for a complete understanding of protein structure and function.

### Global dependency

Global dependency refers to the long-range interactions between distant amino acids in a protein sequence, which are crucial for determining the overall structure and function of the protein. These dependencies allow for the understanding of how distant regions of a sequence influence one another, playing a significant role in processes like protein folding and stability. Transformer-based models are particularly effective at capturing global dependencies by using a self-attention mechanism (Vaswani et al., 2017), which enables them to analyze all parts of the sequence simultaneously. This mechanism allows the model to understand how distant amino acids interact, providing a more comprehensive view of the sequence and improving predictions of protein behavior.

ProtTrans is the first study to comprehensively compare different large transformer models for protein sequence representation (Elnaggar et al., 2021). The study compares a series of transformer models with ProtVec (Asgari and Mofrad, 2015) and SeqVec (Heinzinger et al., 2019), demonstrating that transformers significantly outperform these earlier methods, particularly in tasks such as secondary structure prediction. Learning the internal representations from large-scale protein data, Rives et al. (2021) propose a bidirectional transformer (ESM-1b) to model protein sequences, capturing both structural and functional features for improved performance on various protein-related tasks. Unlike models that rely solely on individual sequences, MSA Transformer is the first transformer-based model that directly models input sequences using multiple sequence alignments (MSAs) (Rao et al., 2021), capturing evolutionary relationships and sequence dependencies through the transfor-

mer architecture and achieving better performance. Though MSA provides extended information, it also brings more computation and storage costs. Another study models protein sequences using a supervised ESM-1b (s-ESM-1b), which generates sequence tokens for structure prediction, leveraging transformers to capture sequence dependencies without the need for multiple sequence alignments (Wang et al., 2022). Moreover, due to the scalability of transformer models, many protein foundational models (Lv et al., 2024; Madani et al., 2023) are built on this architecture, benefiting from its ability to process large datasets efficiently.

Despite the current dominance of the global dependency paradigm, local dependency remains an integral part of many models. For instance, in the aforementioned global dependency models, the process of converting amino acids into numerical input for models, known as tokenization, inherently captures local dependencies. A typical example is the  $k$ -mer tokenization approach, which considers  $k$  consecutive amino acids as a single unit, emphasizing the local sequence relationships. Furthermore, methods like byte-pair encoding extend this concept by incorporating statistical local information (Sennrich et al., 2016), identifying frequently occurring patterns of amino acids within sequences. These techniques ensure that local dependencies are not only preserved but also effectively utilized to complement the broader global context, enhancing the overall predictive power of the models.

### Structure representation

Protein structure refers to the three-dimensional arrangement of amino acids within a protein, including its secondary, tertiary, and quaternary structures. Accurate modeling of protein structure is essential for tasks like protein design and drug discovery, as structure directly influences protein behavior. Protein structure modeling benefits from two key approaches: coordinate matrices and graph representations (Figure 2B). Coordinate matrices capture the precise atomic positions, providing detailed input for AI models to predict protein behavior at the atomic level. Graph representations, on the other hand, model the spatial and functional relationships between amino acids, making them particularly useful for capturing non-linear interactions and long-range dependencies within proteins. These complementary methods enable AI models to effectively predict protein structures, interactions, and functions, driving advancements in protein design and other applications.

### Coordinate matrix

The coordinate matrix approach records the precise 3D spatial coordinates of each atom in a protein structure. By storing atomic positions in a matrix, this method provides detailed input for AI models to predict protein behavior, interaction sites, and conformational changes with high accuracy. Many protein datasets are structured as coordinate matrices, such as PDB-101 (Zardecki et al., 2022). Thus the input of models is always coordinate matrices of various atoms. For example, RFDiffusion represents residues using alpha-carbon (C $\alpha$ ) coordinates and N-C $\alpha$ -C rigid orientations and utilizes a diffusion-based generative model for *de novo* protein design (Watson et al., 2023). These coordinates and orientations are represented as the input and output matrices to model protein structures. Additionally, Lan et al. (2024) combine alpha-carbon (C $\alpha$ ) 3D coordinates with amino acid sequences and receptor-ligand indexes, guiding the design and prediction of protein mutants with altered binding affinities. AlphaFold3 also uses a coordinate matrix as part of its modeling approach during the diffusion-based prediction process (Abramson et al., 2024).

### Graph representation

Proteins involve both local and long-range interactions, such as spatial, biochemical, and functional relationships. Graph representations capture protein information by modeling amino acids or atoms as nodes and their interactions or spatial relationships as edges. Graph neural networks (GNNs) are usually applied to capture both short- and long-range dependencies based on graph representations (Zhou et al., 2020), as GNNs iteratively aggregate information across the entire graph. For instance, RoseTTAFold models amino acids as nodes and their interactions as edges (Baek et al., 2021). This graph helps capture local structural relationships, which are integrated with other information to improve overall protein structure prediction. Similarly, Ingraham et al. (2023) regard residues as nodes and use graph neural networks to process spatial and interaction data. Zhang et al. (2022) also model protein structures as graphs where residues are represented as nodes, and their spatial and sequential interactions are captured through edges. Réau et al. (2023) take Protein Data Bank (PDB) 3D coordinate files as input and model protein-protein interactions as graphs, facilitating learning of specific interaction patterns.

### Protein property representation

Protein properties, including physicochemical and evolutionary information, are essential for understanding how proteins interact, their stability under different conditions, and their roles in biological processes such as catalysis and signaling. By combining insights from physicochemical data and evolutionary information, AI models can make more reliable predictions of protein function and stability, providing a deeper understanding of how proteins operate in diverse biological contexts. This representation modeling involves capturing both physical interactions at a molecular level and evolutionary patterns that guide protein behavior.

### Physicochemical information

Physicochemical information encompasses the fundamental physical and chemical properties of proteins, such as hydrophobicity, charge, and molecular size. These properties play a

crucial role in how proteins behave in various environments, such as force energy contributing to structural stability and hydrophobicity predicting interactions and solubility. To model these properties in AI systems, they are represented through functions or scores that capture their effects. For example, force and energy are modeled using energy functions, with gradients representing forces on each atom, ensuring generated protein structures adhere to physical stability principles (Wang et al., 2024). Similarly, hydrophobicity is quantified using the Kyte-Doolittle scale (Kyte and Doolittle, 1982) and integrated into AI models (Villegas-Morcillo et al., 2024), guiding predictions of solubility and stability. These representations enable AI models to incorporate essential physicochemical properties for accurate protein structure generation.

### Evolutional information

Evolutionary information refers to the data that reflect the relationships and changes among protein sequences over time, which is important for understanding the functional and structural similarities of homologous proteins. MSA aligns multiple biomolecules' sequences to identify regions of similarity that may indicate functional, structural, or evolutionary relationships. In AI models, MSA is typically represented as a matrix, where each row corresponds to an aligned protein sequence, and each column represents positions in the sequences that are compared across different proteins. MSA was first implemented using dynamic programming algorithms like ClustalW (Thompson et al., 1994) and MUSCLE (Edgar, 2004) to identify conserved regions in sequences. Then, MSA was integrated with deep learning studies such as DeepMSA (Zhang et al., 2020). This integration allows the model to better capture evolutionary relationships, which improves the overall performance of protein structure prediction tasks. With the rise of transformer, MSA Transformer innovatively integrates MSA with transformer architecture, allowing the model to directly process aligned sequences and effectively capture the intricate evolutionary relationships and dependencies within the data (Rao et al., 2021). Many important studies, including RoseTTAFold (Baek et al., 2021) and AlphaFold (Jumper et al., 2021), also adopt MSA as important modules for advanced prediction accuracy. However, MSA's heavy reliance on data leads to high computational and memory demands. As a result, AlphaFold3 focuses on reducing dependency on MSA by using single-sequence predictions while still incorporating evolutionary information (Abramson et al., 2024). It reduces the amount of MSA by replacing the AlphaFold2 Evoformer with the simpler pairformer module. Furthermore, MSA processing is de-emphasized, with a much smaller and simpler MSA embedding block within the model.

### Generating proteins

The generation of protein sequences that exhibit desired functions is the key to AI-assisted protein engineering. Generative AI has the potential to reshape protein design by enabling direct sequence generation as well as structure-informed sequence design. Direct sequence generation focuses on producing new protein sequences with desired characteristics, while structure-informed design starts from a target structure and designs sequences that fold into it. Both approaches offer complementary insights into protein functionality (Figure 3). In

this section, we introduce the techniques for these two types of sequence generation.

## Generating sequences

Although progress has been made in mining existing protein sequences, the gap between theoretical and known sequence spaces remains vast. The theoretical protein sequence space, estimated to be around  $20^{100}$ , comes from the possible combinations of 20 amino acids in proteins averaging 100 residues in length. In contrast, the number of known biological protein sequences is only about  $20^6$ , based on experimental data and predictions stored in databases like UniProt and GenBank. This vast difference highlights the necessity of directly generating proteins with specific functions and properties to explore the full potential of protein design. AI models can accelerate the discovery of new functional protein sequences by learning to construct the latent knowledge and relationships within protein sequences. In this section, we will introduce unconditional sequence generation and conditional sequence generation (Figure 3A).

### Unconditional generation

For unconditional generation, models generate protein sequences by focusing exclusively on patterns found in known sequence data and without external information. For instance, Riesselman et al. (2017) capture complex interactions within protein sequences using unsupervised learning, enabling the prediction of mutation effects by uncovering latent evolutionary constraints. ProGen enables protein sequence generation with predictable functions by leveraging deep-learning language models trained on 280 million sequences, augmented with control tags and fine-tuning for diverse protein families (Madani et al., 2023). Wan et al. (2024) train on extinct organism proteomes to learn patterns associated with antimicrobial activity, and the model was further employed to generate and predict new peptide sequences with potential antibiotic properties. Ruffolo et al. (2024) train a large language model based on large-scale protein sequences and generate millions of CRISPR-Cas proteins. With selection and optimization, a highly functional synthetic gene editor, OpenCRISPR-1, is designed. Further than protein sequences, Nguyen et al. (2024) train on whole-genome sequences, allowing the capture of intricate patterns of genetic information encoding and the generation of a novel CRISPR system. These approaches demonstrate how sequence-only models can uncover hidden relationships in sequence data and how they can be applied to generate functionally relevant protein sequences.

### Conditional generation

Conditional generation models produce protein sequences based on specific function constraints, such as desired mutation effects, binding affinity, or structural features. Compared with directly generating unconditional sequences, these models can generate sequences that are tailored to meet particular requirements by incorporating constraints, making them powerful tools for targeted protein design. For example, EVOLVEpro integrates protein language models and regression techniques on protein mutation datasets to address challenges in directed evolution (Jiang et al., 2025), achieving up to 100-fold improvements in desired properties with minimal experimental data. Wait et al.

(2024) design fluorescent calcium indicators by training models on mutation data, predict the impact of each mutation on dynamic range and response kinetics, and then use these predictions to iteratively generate and optimize new protein sequences to achieve enhanced performance. Frey et al. (2023) introduce Discrete Walk-Jump Sampling (dWJS), which uses energy-based models to generate and refine antibody sequences with small (walk) and larger (jump) mutations. Meanwhile, CoVES uses structural microenvironment data to predict how combinatorial sequence mutations affect protein function, guiding the design of optimized protein variants based on local structural features (Ding et al., 2024). DeepDirect introduces binding affinity as the condition on the other hand (Lan et al., 2024). It leverages a Generative Adversarial Network (GAN) (Arjovsky et al., 2017), where one model creates fake data and another tries to detect it, helping the generator improve and produce more realistic results. DeepDirect can predict and optimize mutations that increase or decrease binding affinity, thus generating protein sequences with specified binding affinity changes. Structural information can also serve as conditions in protein sequence generation. Anishchenko et al. (2021) employ deep network hallucination to conditionally generate new protein sequences by iteratively optimizing them to match specific structural distributions, enabling the *de novo* design of novel, stable, and functional proteins.

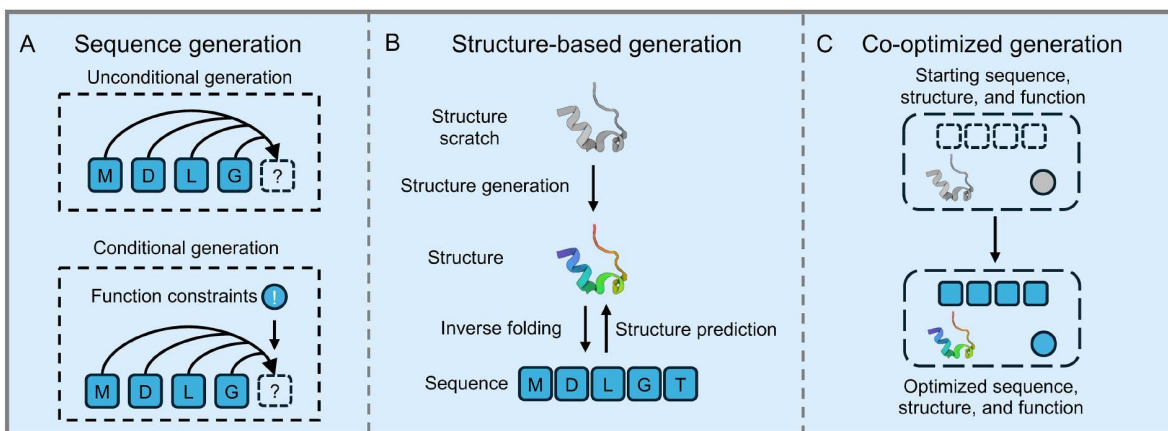
## Structure-based generation

In protein design, the correct folding of a sequence into its functional structure is essential for understanding and manipulating protein behavior, and misfolding can lead to the loss of activity or disease. Therefore, integrating structural information into protein sequence design is crucial to guide sequences toward proper folding and function.

In this section, we will explore three key components of structure-based generation (Figure 3B). First, inverse folding focuses on designing sequences that fold into a given structure, offering a way to reverse-engineer protein sequences based on structural constraints. Next, structure prediction ensures that generated sequences can reliably fold into their intended structures, addressing the challenge of sequence-to-structure mapping. Finally, structure generation emphasizes creating new structural forms, which are then used as blueprints for designing corresponding sequences. Together, these methods provide a comprehensive framework for guiding protein design using structural information, helping to improve both the accuracy and functional relevance of the generated proteins.

### Inverse folding

Inverse folding refers to identifying protein sequences that can fold into a predefined three-dimensional structure. The correct protein folding confers precise functional and structural attributes; thus the design of corresponding protein sequences is also important. Inverse folding bridges sequence design and structural biology, enabling the creation of sequences that are specifically tailored to achieve desired structural configurations. The emergence of structure prediction models like RoseTTAFold (Baek et al., 2021), AlphaFold2 (Jumper et al., 2021), RoseTTAFold All-Atom (Krishna et al., 2024), and AlphaFold3 (Abramson et al., 2024) has generated vast amounts of accurate protein structures, creating new opportunities for inverse folding models.



**Figure 3.** The overview of protein generation approaches. A, Unconditional generation designs sequences by capturing patterns found in known sequence data; conditional generation generates protein sequences relied not only on existing sequence data, but also on other function constraints. B, Structure-based generation introduces structure information in the generation process. Structure generation generates a desired functional protein structure from a random initialized scratch. Inverse folding generates protein sequences that can properly fold into the desired functional structure. The structure prediction can predict the folding structure of given sequences, helping examine the folding of designed sequences. C, Based on sequence, structure data, and function constraints, co-optimized models can simultaneously optimize both sequential and structural protein designs.

CarbonDesign addresses inverse folding by predicting protein sequences for given structures inspired by AlphaFold techniques (Ren et al., 2024), using a module to learn backbone structure representations and a Markov random fields model to decode sequences. ProteinMPNN designs sequences to match specific protein structures by using structured message-passing to capture dependencies between residues (Dauparas et al., 2022), improving the alignment of sequences to their target structures and rescuing previously failed designs made with Rosetta or AlphaFold. Hsu et al. (2022) use large-scale predicted structures from AlphaFold (Jumper et al., 2021) to design protein sequences, enhancing sequence recovery by integrating geometric vector perceptrons with transformer models for accurate fixed-backbone protein design.

### Structure prediction

Structure prediction models aim to determine a protein's three-dimensional fold directly from its amino acid sequence. This folding process is governed by complex interactions between residues, and errors in folding can lead to the loss of function or disease. AI models handle this complexity by processing large datasets and identifying the patterns that dictate protein folding, allowing for faster and more accurate predictions. As early as 2004, Rosetta was able to predict protein structures by employing a combination of energy-based scoring functions and algorithms that simulate folding processes (Rohl et al., 2004), generating and evaluating numerous conformations to identify the most stable structure. Now, advanced and important structure prediction models such as RoseTTAFold (Baek et al., 2021), AlphaFold2 (Jumper et al., 2021), RoseTTAFold All-Atom (Krishna et al., 2024), and AlphaFold3 (Abramson et al., 2024) predict the three-dimensional structure of proteins from their amino acid sequences using deep learning. These models leverage large datasets of protein structures to learn the complex relationships between sequence and structure, achieving high accuracy in predicting protein folds, which significantly accelerates structural biology research and enables new insights into protein function and design. Building on the success of AlphaFold2 and RoseTTAFold, RoseTTAFold All-Atom refines

predictions at the side-chain and atomic levels, offering a more detailed and accurate representation of protein structure. In contrast, AlphaFold3 further advances the field by improving predictions for multi-chain complexes and protein-protein interactions, enabling more effective analysis of large and complex systems. Protein property information such as forces and evolutionary patterns can also aid the generation. ConfDiff generates diverse and physically plausible protein conformations from a given sequence (Wang et al., 2024), using a force-guided SE(3) diffusion process (Yim et al., 2023) constrained by the sequence to ensure alignment with the Boltzmann distribution and structural fidelity. Lin et al. (2023) present a large language model that predicts atomic-level protein structures directly from sequence data by learning evolutionary patterns across millions of sequences, enabling rapid, high-resolution structure predictions.

### Structure generation

Structure generation involves designing new protein structures before determining the corresponding sequences. This approach enables the exploration of novel protein shapes, allowing further generation of sequences that fold into these target structures. Starting with a structure scratch, researchers could gain functional and stable desired structure through structure generation. For example, RFDiffusion starts from random noisy inputs and gradually refines them into functional proteins (Watson et al., 2023). The model learns how to reverse noise and recover realistic protein shapes, ensuring that the final designs are stable and meet functional requirements. Further advancement employs RFDiffusion to design high-affinity binders for helical peptides (Vázquez Torres et al., 2024), enabling *de novo* creation of binders with picomolar affinities based on learned structural and interaction patterns. Another study uses RFDiffusion to design *de novo* proteins specifically engineered to neutralize snake venom toxins (Vázquez Torres et al., 2025). It generates protein backbones optimized for binding neurotoxins and cytotoxins, meeting the precise requirement of high affinity and stability. Further studies, such as Chroma (Ingraham et al., 2023), generate protein structures from random noise using a

diffusion model conditioned on properties like symmetry and shape, offering more customization than RFDiffusion.

### Co-optimized sequence-structure generation

Generating sequences and structures independently can lead to mismatches, as sequence determines structure, and structure impacts function. This interdependence requires an integrated approach to ensure that both elements work together for stable and functional proteins. AI models now tackle this interdependence by learning the relationship between sequence and structure, allowing for the simultaneous refinement of both (Figure 3C). For example, ProteinGenerator integrates RoseTTAFold (Baek et al., 2021) and a diffusion model (Ho et al., 2020) to co-design protein sequences and structures (Lisanza et al., 2023). This ensures that the designed proteins meet specific requirements such as secondary structures or functional motifs, enhancing precision in protein design for targeted applications. Wang et al. (2022) also use RoseTTAFold to simultaneously design protein sequences and structures that scaffold functional sites, enabling the creation of novel, functional proteins by optimizing both sequence and structure in an integrated manner. Jiang et al. (2024) co-optimize sequence and structure by using AlphaFold (Jumper et al., 2021; Jumper and Hassabis, 2022) to predict structural outcomes from sequence modifications, ensuring stability in loop regions.

### Evaluating protein properties

In the protein design workflow, evaluating the properties of designed proteins is a critical step to ensure that they exhibit desired characteristics such as stability, binding affinity, and activity. These properties are often closely tied to how proteins interact with other molecules, with underlying physicochemical features driving these interactions. Traditionally, validating these hypotheses requires time-consuming and resource-intensive experiments. AI offers a more efficient alternative by predicting properties and interactions from large datasets, reducing the need for extensive lab work. This section explores how AI enhances predictions of protein functions and interactions, leveraging their interconnected nature.

### Functional prediction

Predicting protein functions, such as enzyme activity, is crucial for understanding biological roles but difficult due to the complexity of sequence-function relationships. AI models help by efficiently analyzing large datasets to uncover patterns that might be missed through traditional approaches, offering faster and more accurate predictions. The training and evaluation processes are based on existing or collected labeled protein datasets (Lovell et al., 2003; Price et al., 2018). For example, DeepFRI uses graph convolutional networks (GCNs) to predict various protein functions (Gligorijević et al., 2021), focusing on tasks such as gene ontology annotations (molecular function, biological process, and cellular component) and enzyme activity classification through enzyme commission (EC) numbers, leveraging both protein structural and sequence data for accurate predictions. CLEAN is also designed to accurately predict enzyme functions, particularly EC numbers, within label-constrained scenarios (Yu et al., 2023). Rita trains on over 280 million

protein sequences and is effective in enzyme function prediction and mutation effect prediction (Hesslow et al., 2022). Some physicochemical properties can also be predicted by models for protein design. RaSP predicts protein stability by calculating changes in free energy caused by mutations (Blaabjerg et al., 2023), where an increase in free energy indicates destabilization. To be noticed, the emergence of generative models such as AlphaFold brings a large quantity of predicted structure data, and thus supplies more data and improves the prediction performance of protein function (Ma et al., 2022).

### Interaction prediction

In protein research, understanding interactions is crucial, as many protein functions depend on how proteins interact with other molecules. These interactions, whether between proteins, ligands, or substrates, are key drivers of biological functions. Traditionally, predicting interactions requires resource-intensive experiments, but AI now enables faster, large-scale predictions, providing insights into both interactions and the resulting functions. For example, Liu et al. (2024) predict deubiquitinase-substrate interactions using evolutionary data, with model performance evaluated by accurately identifying the existence of critical protein-protein relationships. Similarly, Lei et al. (2021) employ deep learning to capture peptide-protein interactions, identifying binding residues and evaluating binary interactions using precision and recall scores. In enzyme-substrate interaction modeling, Xu et al. (2022) introduce a substrate-encoding approach, showing significant improvements in predictive performance. Furthermore, Rao et al. (2024) integrate multi-scale information, evaluating comprehensive interaction predictions across protein-protein, drug-protein, and drug-drug interactions. Additionally, energy landscape simulations complement AI models by providing insights into stable conformations and energy barriers that affect interactions (Neelamraju et al., 2020), offering a broader understanding of protein behavior (McBride and Tlustý, 2024).

### Comprehensive prediction

Further than specific applications in functional prediction and interaction analysis, large models are trained on large datasets and can capture essential biological patterns, making them highly adaptable to various tasks. Their key advantage lies in their ability to generalize across different applications, allowing researchers to apply them to a wide range of protein-related problems. For example, some studies present how scaling up over a large protein dataset leads to advances in protein representation (Rives et al., 2021; Su et al., 2023; Zhang et al., 2023). These foundational models have been applied to both function and interaction prediction tasks. Another approach highlights the use of codons, rather than protein sequences, to construct large language models (Outeiral and Deane, 2024), which ultimately results in more accurate predictions across tasks like species recognition, prediction of protein, and transcript abundance, or melting point estimation. LucaOne further builds a foundational model based on further biological data like DNA, RNA, and protein data (He et al., 2024), and thus is capable of diverse biological tasks such as taxonomic annotation, subcellular localization, protein stability, and molecular interactions.

An example that underscores the potential of foundational

models in comprehensive prediction is the task of predicting post-translational modifications (PTMs). PTMs play a pivotal role in modulating protein conformations, functionalities, and interactions within cellular contexts. The prediction of PTMs can facilitate the process of studying new proteins. PTMGPT2 employs unsupervised learning and prompt-based fine-tuning to achieve accurate PTM predictions (Shrestha et al., 2024). By identifying sequence motifs and analyzing mutation effects, PTMGPT2 enhances our understanding of PTM-related mechanisms and highlights the potential for foundational models to serve complex predictive tasks, further advancing the design of proteins with tailored functionalities.

## Grand challenges

AI has transformed protein design by enhancing data representation, protein design, and property evaluation. A vast number of protein models have been developed (Table 1). However, key challenges such as balancing MSA and single-sequence approaches, development of high-quality protein datasets, and enhancing model interpretability remain. Looking forward, the integration of AI into wet-lab automation offers great potential, enabling more efficient experimentation and expanding the scope of protein discovery.

## Algorithmic innovation

As AI continues to revolutionize protein research, there are several key algorithmic challenges that need to be addressed for further advancements. One of the central issues is how models handle sequence data. MSA offers the advantage of capturing evolutionary information, helping models identify conserved patterns crucial for function and structure prediction. However, MSA requires vast computational resources for storage and analysis. Single-sequence approaches, while avoiding these demands, lack comparative data and struggle to capture evolutionary signals. Some recent works, such as ESM (Lin et al., 2023), address the computational cost and data access challenges of MSA by using self-supervised or unsupervised learning on large datasets. These models extract sufficient information directly from single sequences, eliminating the need for MSA while maintaining accurate predictions.

While unsupervised and self-supervised learning methods help reduce the need for MSA by extracting informative features from single sequences (Chen et al., 2023; Ferruz et al., 2022; Lin et al., 2023), they still rely on vast amounts of data. It further underscores the importance of high-quality, annotated protein datasets. Much like ImageNet transformed computer vision (Deng et al., 2009), these datasets are essential for training robust AI models, enabling them to generalize well across diverse protein prediction tasks. Curating such datasets, however, remains a significant challenge. An emerging solution is to use AI-generated protein sequences as a data source. The generation results of models such as AlphaFold (Jumper et al., 2021) serve as a training set for model pretraining (Hsu et al., 2022). These generated data offer a flexible and scalable approach to creating diverse datasets for training AI systems, reducing the dependence on labor-intensive data curation processes. While these predictions have proven invaluable, it is important to note that they may reflect biases or limitations inherent in the model, particularly in less-characterized or disordered protein regions.

To effectively leverage these high-quality datasets, it is crucial to consider the methodologies for extracting insights—whether focusing on single data points or exploring relationships among multiple data points. Models focusing on single data points aim to extract extensive internal information from vast datasets (Watson et al., 2023). In contrast, relationship models, such as those that employ graph-based representations (Ingraham et al., 2023), focus more on the connections between substances, emphasizing how protein interactions dictate functional outcomes. The connection between these approaches is akin to the relationship between function prediction and interaction prediction. In function prediction, the goal is often to infer properties from a single protein, which may superficially seem like a discrete task. However, this process inherently involves understanding interactions, as the function of a protein is deeply intertwined with its molecular interactions within a biological context. Conversely, interaction prediction explicitly targets the relationships between proteins, revealing how these interactions contribute to overall functionality. This comparison illustrates that while single data point analyses focus on the individual features of data, they must also consider the relationship dynamics that influence those characteristics. Similarly, relational approaches emphasize the connections among data points, yet the outcomes of these relationships are ultimately reflective of their individual characteristics. By integrating insights from both methodologies, researchers can develop a more comprehensive understanding of protein behavior, enhancing performance across various tasks.

Finally, model interpretability is becoming an increasingly important area of focus. As AI models become more complex, understanding the underlying mechanisms driving their predictions becomes critical. Improving interpretability allows researchers to trust AI predictions more fully, ensuring that the insights derived are not only accurate but also explainable. This will play a key role in integrating AI into broader biological research and applications, particularly in areas where high confidence and transparency are essential.

## Laboratory automation

Laboratory automation, especially in wet-lab settings, is transforming protein research by reducing manual intervention, enhancing reproducibility and improving consistency and throughput. Traditional experiments are often slow, labor-intensive, and prone to variability, making it difficult to scale and optimize workflows. Though few studies focus specifically on protein wet-lab automation, automation workflows from biological and chemical research provide valuable examples. CRISPR-GPT (Huang et al., 2024), for instance, automates gene-editing design, while Coscientist integrates AI with robotic systems to autonomously execute complex chemical tasks (Boiko et al., 2023). A-Lab also demonstrates significant efficiency improvements by autonomously synthesizing novel compounds (Szymanski et al., 2023). These advancements set a precedent for protein research automation.

Drawing from the successful applications in other fields, the future of laboratory automation in protein design holds great promise. However, automating wet-lab experiments in protein design still presents unique challenges compared with other fields. One significant difficulty lies in the complexity and variability of biological systems. Proteins have intricate struc-

**Table 1.** The summary of protein models, including the category of the trained data and the brief description of the models

| Model name           | Dataset type                                                                                              | Description                                                                                                                                                                                                                       | Ref                         |
|----------------------|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| AlphaFold3           | Protein structural data                                                                                   | AlphaFold3 builds on its predecessor by integrating advanced modeling techniques, improving accuracy and scalability in predicting complex protein structures and enabling better handling of protein families with diverse folds | (Abramson et al., 2024)     |
| RaSP                 | Experimental stability data, protein structures, and mutation information across various protein families | A novel computational framework for predicting the effects of mutations on protein stability, providing insights into the relationship between protein structure and function                                                     | (Blaabjerg et al., 2023)    |
| ProteinMPNN          | Protein sequential and backbone data                                                                      | A deep learning based protein sequence design method that shows outstanding performance in both <i>in silico</i> and experimental tests                                                                                           | (Dauparas et al., 2022)     |
| ProtTrans            | Protein sequential data                                                                                   | A trained model shows that scaling large language models for protein sequences improves prediction accuracy of protein properties                                                                                                 | (Elnaggar et al., 2021)     |
| ProtGPT2             | Protein sequential data                                                                                   | A language model based on protein space that creates novel protein sequences in accordance with the principles observed in naturally occurring proteins                                                                           | (Ferruz et al., 2022)       |
| DeepFRI              | Protein structural data                                                                                   | A graph convolutional network designed to predict protein functions by utilizing sequence features derived from a protein language model alongside protein structural information                                                 | (Gligorijević et al., 2021) |
| LucaOne              | DNA and RNA sequential data, protein structural and sequential data                                       | A biological language model that integrates genetic and proteomic data, aiming at universal bioinformatics applications                                                                                                           | (He et al., 2024)           |
| Rita                 | Protein sequential data                                                                                   | A protein sequence autoregressive generative model for protein design                                                                                                                                                             | (Hesslow et al., 2022)      |
| Chroma               | Protein structural, sequential, and functional data                                                       | A generative model for proteins and complexes that allows conditioning to achieve specific properties and functions                                                                                                               | (Ingraham et al., 2023)     |
| AlphaFold            | Protein structural data                                                                                   | AlphaFold significantly advances protein structure prediction accuracy from amino acid sequences using deep learning                                                                                                              | (Jumper et al., 2021)       |
| RoseTTAFold All-Atom | Protein structural data                                                                                   | A model enables accurate modeling of complex biological assemblies including proteins and small molecules, and predicting of covalent modifications and interactions                                                              | (Krishna et al., 2024)      |
| CAMP                 | Protein-peptide complex data and drug-target interaction data                                             | A model to predict peptide-protein interactions and identify binding residues                                                                                                                                                     | (Lei et al., 2021)          |
| ESM-2                | Protein sequential data                                                                                   | The first large-scale structural characterization of metagenomic proteins                                                                                                                                                         | (Lin et al., 2023)          |
| ProteinGenerator     | Protein sequential data                                                                                   | A sequence space diffusion model that simultaneously generates protein sequences and structures                                                                                                                                   | (Lisanza et al., 2023)      |
| ProLLaMA             | Protein instruction data developed on sequential data                                                     | A protein language model that can simultaneously handle multiple protein language processing tasks                                                                                                                                | (Lv et al., 2024)           |
| ProGen               | Protein sequential data                                                                                   | A language model that generates protein sequences with predictable functions across large protein families                                                                                                                        | (Madani et al., 2023)       |
| Evo                  | Genome sequential data                                                                                    | A genomic foundation model that facilitates prediction and generation tasks spanning from the molecular level to the genome scale                                                                                                 | (Nguyen et al., 2024)       |
| CaLM                 | Codon sequential data                                                                                     | A large language model trains on codons and provides high-quality representations that is capable on a variety of protein tasks                                                                                                   | (Outeiral and Deane, 2024)  |
| MSA transformer      | Protein sequential data                                                                                   | A protein language model leverages multiple sequence alignments to achieve superior performance in unsupervised structure learning and great parameter efficiency                                                                 | (Rao et al., 2021)          |
| DeepSequence         | Protein and genome sequential data                                                                        | A probabilistic model for sequence families, predicting the effects of mutations                                                                                                                                                  | (Riesselman et al., 2017)   |
| ESM                  | Protein sequential data                                                                                   | A deep contextual language model that enables prediction of mutational effect, secondary structure, and features for long-range contact                                                                                           | (Rives et al., 2021)        |
| OpenCRISPR-1         | Genome data                                                                                               | An AI-designed gene editor that enables precision editing of the human genome                                                                                                                                                     | (Ruffolo et al., 2024)      |
| Saprot               | Protein structural and sequential data                                                                    | A structure-aware protein language model that integrates residue and structure tokens, achieving superior performance across ten significant downstream tasks                                                                     | (Su et al., 2023)           |
| RFdiffusion          | Protein structural and sequential data                                                                    | A diffusion-based generative model enabling high-performance protein design, such as <i>de novo</i> binders and symmetric architectures                                                                                           | (Watson et al., 2023)       |
| APEX                 | Peptide structural and sequential data                                                                    | A model facilitates discovering novel antibiotic peptides from extinct organisms                                                                                                                                                  | (Wan et al., 2024)          |
| trRosettaX-Single    | Protein sequential data                                                                                   | A transformer-based model predicts protein structures from single sequences                                                                                                                                                       | (Wang et al., 2022)         |
| ConfDiff             | Protein structural data                                                                                   | A force-guided diffusion model that generates diverse and accurate protein conformations by incorporating physical knowledge into the generation process                                                                          | (Wang et al., 2024)         |
| CLEAN                | Protein sequential data                                                                                   | A contrastive learning-based model that improves enzyme function annotation by accurately assigning enzyme commission (EC) numbers                                                                                                | (Yu et al., 2023)           |
| GearNet              | Protein structural data                                                                                   | A structure-based encoder utilizes a geometry-aware relational graph neural network and improves performance on property prediction tasks                                                                                         | (Zhang et al., 2022)        |
| EvolvePro            | Protein mutational data                                                                                   | A few-shot active learning framework that integrates protein language models and regression to address challenges in directed protein evolution                                                                                   | (Jiang et al., 2025)        |
| PTMGPT2              | Protein modification data                                                                                 | An interpretable protein language model that utilizes prompt-based fine-tuning for post-translational modifications                                                                                                               | (Shrestha et al., 2024)     |

tures and dynamic behaviors that can be highly sensitive to changes in environmental conditions, which makes experimental standardization harder. Additionally, real-time adjustments are often required in protein purification, expression, and folding processes, these are tasks that current automated systems struggle to handle flexibly. Furthermore, replicating the cellular environment for accurate protein interactions adds another layer of complexity, as these conditions are difficult to simulate outside of living systems.

## Conclusion

AI has revolutionized protein design by significantly improving data representation, protein design, and property evaluation, leading to faster discoveries and more precise predictions. The ability of AI to process vast datasets, model complex protein structures, and generate novel designs with high accuracy demonstrates its immense potential in transforming the field. However, challenges such as balancing MSA and single-sequence approaches, the development of high-quality datasets, data analysis approaches, and improving model interpretability remain. Looking ahead, the integration of AI into wet-lab automation holds great promise for making experimentation more efficient and scalable, further accelerating the pace of protein research and discovery.

## Compliance and ethics

The authors declare that they have no conflict of interest.

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